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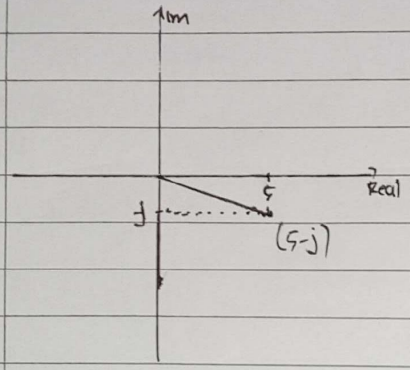
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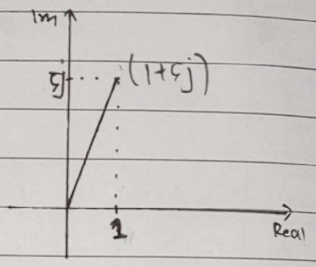
Latihan 3

1. Deket bilangan kompleks : $z_1 = 3+j2$, $z_2 = 2-j3$
Hitung z_1+z_2 dan z_1-z_2 dan gambarkan dalam diagram Argand

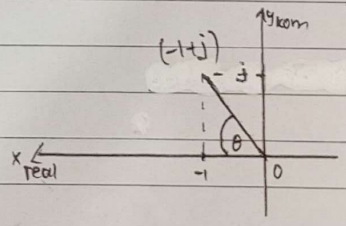
a. $= z_1+z_2$
$= 3+j2+2-j3$
$= 5-j$

b. $= z_1-z_2$
$= 3+j2-(2-j3)$
$= 1+j5$





2. Diketahui bilangan kompleks bentuk Cratesian : a. $-1+j$ b.
Ubahlah ke bentuk Polar dan gambarkan
$z = -1+j$ kuadran II dengan • $r = \sqrt{(-1)^2+(1)^2} = \sqrt{2}$ dan

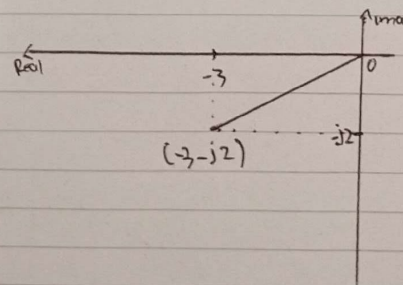


• $\theta = \text{arc tgn}\left(\frac{1}{1}\right) = 45^0$
• jadi $180^0 - 45 = 135^0$
• Bentuk polarnya
$z = \sqrt{2}(\cos 135^0 + j\sin 135^0)$
$z = \sqrt{2} \angle 135^0$

3. Nyatakan dalam bentuk Cartesian/rectangular
a. $z = 32\left(\cos\frac{3\pi}{4} + j\sin\frac{3\pi}{4}\right)$
$z = 32(\cos 135^0 + j\sin 135^0)$
$= 32\left(-\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right)$
$= -16\sqrt{2} + j16\sqrt{2}$
$= -22.63 + j22.63$

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b. $-3-j2$ kuadran III



• $r = \sqrt{(-3)^2+(-2)^2}$
$= \sqrt{13}$
• $\theta = \text{arc tgn}\left(\frac{-2}{-3}\right)$
$= \text{arc tgn}\left(\frac{2}{3}\right) \Rightarrow 33.69^0$
• k III $= 180^0 + 33.69^0$
$= 213.69^0$
• Bentuk polarnya
$z = \sqrt{13}(\cos 213.69^0 + j\sin 213.69^0)$
$z = \sqrt{13} \angle 213.69^0$

$z = c \angle \theta$ (cos → real, sin → imaginer)

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b $z = 3\angle -60$

$a = 3\cos(-60^\circ)$ $b = 3\sin(-60^\circ)$ jadi $z = 1{,}5 - j2{,}598$

$= 3\left(\frac{1}{2}\right)$ $= 3\left(-\frac{1}{2}\sqrt{3}\right)$

$= 3/2$ $= -\frac{3\sqrt{3}}{2}$

$= -2{,}598$

4. Oprasikan perkalian berikut dalam bentuk cartesian dan polar

c) $(-2+j3)(-1-j2)$

$= 2 + 4j - j3 - j^2 6$

$= -j^2 6 + j + 2$

$= -(\sqrt{-1})^2 6 + j + 2$

$= 6 + 2 + j$

$= 8 + j$

- Bentuk polar

- $r = \sqrt{8^2 + 1^2}$

$= \sqrt{65}$

- $\theta = \arctan\left(\frac{1}{8}\right)$

$= 7{,}125^\circ$

- $z = \sqrt{65}\angle 7{,}125^\circ$

b) $(2\sqrt{3} + j2)(2 - j2\sqrt{3})$

$= 4\sqrt{3} - j12 + j4 - j^2 4\sqrt{3}$

$= 4\sqrt{3} - j8 - (\sqrt{-1})^2 4\sqrt{3}$

$= 4\sqrt{3} + 4\sqrt{3} - j8$

$= 8\sqrt{3} - j8 \rightarrow$ kIV

* Polar

- $r = \sqrt{(8\sqrt{3})^2 + (-8)^2}$

$= \sqrt{192 + 64}$

$= \sqrt{256} \Rightarrow 16$

- $\theta = \arctan\left(\frac{-8}{8\sqrt{3}}\right)$

$= \arctan\left(-\frac{\sqrt{3}}{3}\right)$

$= -30^\circ \rightarrow 360^\circ - 30 = 330^\circ$

- Polar $z = 16\angle 330^\circ$

5. Bilangan kompleks dibawah ubah dalam cartesian dan polar

a) $= \frac{3 + j4}{1 - j2}$

$= \frac{3 + j4}{1 - j2} \times \frac{1 + j2}{1 \times j^2}$

$= \frac{3 + j6 + j4 + j^2 8}{1^2 + 2^2} = \frac{3 + j10 - 8}{5} = \frac{-5 + j10}{5} = -1 + j2$

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• Bentuk polarnya $z = -1 + j2 \rightarrow$ kII

$r = \sqrt{(-1)^2 + 2^2}$

$= \sqrt{5}$

$\theta = \tan^{-1}\left(\frac{2}{-1}\right)$

$= \tan^{-1}(-2)$

$= -63.435$

karena kII maka

$= 180 - 63.435$

$= 116.565$

* Jadi bentuk polarnya

$z = \sqrt{5} \angle 116.565^\circ$

b. $\frac{1}{2+j3} + \frac{1}{2-j3} = \frac{2-j3+2+j3}{2^2-(j3)^2} = \frac{4}{4-j^2 9}$

$= \frac{4}{4+9} = \frac{4}{13} + j0 =$

* bentuk polarnya

$r = \sqrt{(4/13)^2 + 0^2}$

$= 4/13 = 0{,}307$

$\theta = \tan^{-1}\left(\frac{0}{0{,}307}\right)$

$= \tan^{-1}(0^\circ)$

$= 0^\circ$

* Jadi bentuk polarnya

$z = 0{,}307 \angle 0^\circ$

12. Operasikan dalam bentuk polar

ⓐ $= 3\angle 23^\circ \times 4\angle 46^\circ$

$= 12 \angle 23^\circ + 46^\circ$

$= 12 \angle 69^\circ$

ⓑ $= 8\angle 110^\circ \times 5\angle -45^\circ$

$= 40 \angle 110^\circ - 45^\circ$

$= 40 \angle 65^\circ$

ⓒ $= 2\angle -21^\circ \times 6\angle -58^\circ$

$= 12 \angle -21^\circ + (-58^\circ)$

$= 12 \angle -79$

ⓓ $\frac{7\angle 20^\circ}{3\angle 50^\circ} \Rightarrow 2{,}33 \angle 20^\circ - 50^\circ$

$= 2{,}33 \angle -30^\circ$

ⓔ $\frac{10\angle -40^\circ}{5\angle 20^\circ} = 2\angle -40^\circ - 20^\circ$

$= 2 \angle -60^\circ$

ⓕ $\frac{3\angle -10^\circ}{6\angle -5^\circ} = 0{,}5 \angle -10^\circ - (-5^\circ)$

$= 0{,}5 \angle -5^\circ$

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- Contoh 1 : Tentukan bagian riil dan imajiner bil kompleks berikut

  a) $4+j3$ b) $4-j3$ c) $j2$

  jawab

  a) $z = 4+j3$; $Re(z) = 4$ dan $Im(z) = 3$

  b) $z = 4-j3$; $Re(z) = 4$ dan $Im(z) = -3$

  c) $z = j2$; $Re(z) = 0$ dan $Im(z) = 2$.

- Contoh 2

  Gambarkan bilangan2 kompleks : $4+j3$, $-3+j2$, $-4-j2$, dan $2-j3$ pada diagram argand

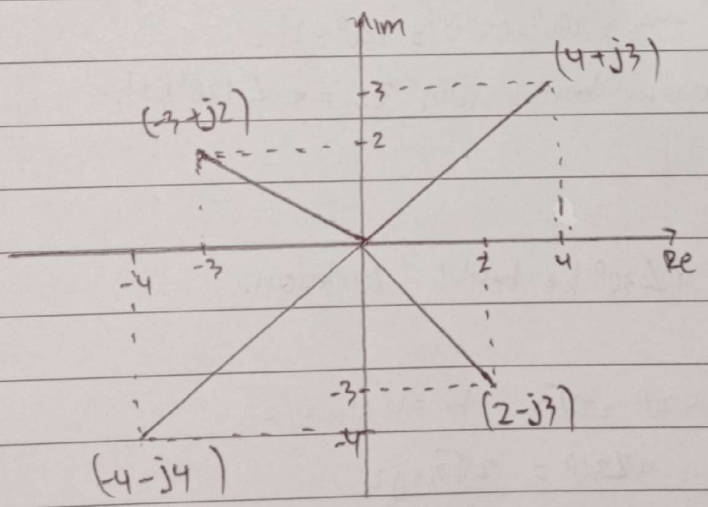
  penyelesaian

  $z = 4+j3$; $Re(z) = 4$ dan $Im(z) = 3$, di kuadran I

  $z = -3+j2$; $Re(z) = -3$ dan $Im(z) = 2$, dan z di kuadran II

  $z = -4-j2$; $Re(z) = -4$ dan $Im(z) = -2$, dan z dikuadran III

  $z = 2-j3$; $Re(z) = 2$ dan $Im(z) = -3$, dan z dikuadran IV



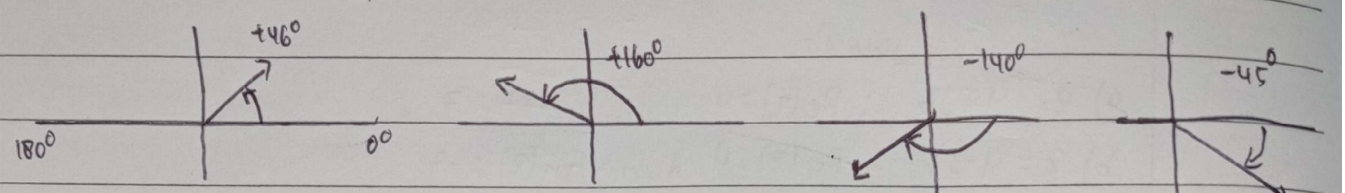
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- Contoh 3

Gambarkan bil kompleks $z = r\angle 46^0$, $z = r\angle 160^0$, $z = r\angle -140^0$, dan $z = r\angle -45^0$



$z = r\angle 46^0$ $z = r\angle 160^0$ $z = r\angle -140^0$ $z = r\angle -45^0$

- Contoh 4

Nyatakan $z_1 = 3 + j4$ dan $z = -3 + j4$ dalam bentuk polar

jawab

- $z_1 = 3 + j4$ di KI dengan $r = \sqrt{3^2 + 4^2} \Rightarrow 5$ dan $\theta = \text{arc tgn } 4/3 = 53^0 8'$

jadi, z_1 dalam bentuk polar adalah $z_1 = 5\angle 53^0 8'$.

- $z_2 = -3 + j4$ berada di kwadran II dengan $r = \sqrt{(-3)^2 + 4^2} \Rightarrow 5$

$\theta = \text{arc tgn } -4/3 = 180^0 - 53^0 8' = 126^0 52'$.

Jadi, z_2 dalam bentuk polar $z_2 = 5\angle 126^0 52'$.

- Contoh 5

Ubahlah $z = 4\angle 30^0$ ke bentuk cartesian.

Penyelesaian

$a = 4\cos 30^0 = 2\sqrt{3}$, $b = 4\sin 30^0 = 2$

jadi, $z = 4\angle 30^0 = 2\sqrt{3} + j2$

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